

Translating Gamification Frameworks for Clinical Mental Health Implementation Lessons and User Research Insight

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Abstract

While ubiquitous computing research has demonstrated the technical feasibility of passive mental health monitoring through smartphone sensing, a critical gap remains between detection capabilities and sustained user engagement with interventions. This paper presents practical lessons learned from implementing gamified behavioral interventions specifically designed for clinical mental health populations, including schizophrenia patients, individuals managing addiction, and postpartum mental health challenges. Through rigorous user research across multiple clinical contexts and deployment of gamification systems supporting FD Approved digital therapeutics, we identify key adaptations necessary when translating established frameworks like Octalysis and the Fogg Behavior Model to mental health contexts. User research in support of our smartphone-first approach to ubiquitous mental health support revealed that social mechanisms consistently disengaged users across clinical populations and must be used with extreme care. Achievement systems require careful calibration to avoid triggering negative responses in vulnerable users, and transparency in reward structures becomes critical for populations already experiencing reduced agency due to their conditions. User testing with schizophrenia patients revealed extreme variability requiring adaptive difficulty scaling that matched individual cognitive capabilities. These findings contribute actionable guidance for researchers developing ubiquitous computing technologies that incorporate game elements for clinical mental health support, emphasizing user-centered adaptation of existing frameworks rather than direct application of general population strategies.

CCS Concepts

• Human-centered computing~Human computer interaction (HCI) • Applied computing~Life and medical sciences • Human-centered computing~Interaction design~Interaction design process and methods • Human-centered computing~Interaction design~Interaction design process and methods~User centered design



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Introduction

Ubiquitous computing technologies have emerged as powerful tools for mental health monitoring, with systems like CrossCheck demonstrating the ability to passively detect behavioral changes in schizophrenia patients through smartphone sensing [1]. However, a fundamental challenge persists: while sensing technologies excel at identifying when clinical intervention is needed, motivating vulnerable populations to engage with recommended interventions remains an unsolved problem in the field.

Personal sensing research has shown that smartphones can collect and analyze data from sensors embedded in the context of daily life with the aim of identifying human behaviors, thoughts, feelings, and traits [2], yet this detection capability does not automatically translate to therapeutic engagement. The promise of just-in-time adaptive interventions (JITAI)s—providing the right type (or amount) of support, at the right time, while eliminating support provision that is not beneficial [3]—offers a compelling framework for bridging this gap through gamification.

When designers attempt to gamify mental health interventions, they typically reach for proven frameworks like Yu-Kai Chou's Octalysis [4] or BJ Fogg's Behavior Model [5,6]. These frameworks have demonstrated success across consumer applications from DuoLingo to Fitbit. However, applying them directly to clinical mental health populations often produces unexpected failures that highlight the need for specialized adaptation approaches.

Through implementation experience spanning multiple clinical contexts—including cognitive remediation games for schizophrenia patients, gamified interventions for postpartum

mental health support, and ethical design consultation for addiction recovery apps—this work identifies specific considerations necessary when translating general gamification frameworks to vulnerable populations. Our smartphone-first approach to ubiquitous mental health support provides practical insights for researchers developing accessible, scalable interventions that can reach clinical populations in real-world settings [9].

Background & Related Work

Gamification Frameworks

The Fogg Behavior Model [5,6] provides a foundational framework for understanding behavior change through three converging elements: Motivation, Ability, and Prompt (B=MAP). The model emphasizes making target behaviors as easy as possible while celebrating completion to wire in new habits through what Fogg calls “shine,” the feeling of success that reinforces behavior.

Octalysis [4] offers a complementary perspective by mapping eight core drives that motivate human behavior: epic meaning, accomplishment, creativity, ownership, social influence, scarcity, unpredictability, and loss avoidance. These drives are categorized as “white hat” (positive, empowering) or “black hat” (manipulative, potentially harmful) techniques.

Ubiquitous Computing for Mental Health

Recent advances in ubiquitous computing have demonstrated significant progress in mental health monitoring through mobile sensing. Personal sensing refers to collecting and analyzing data from sensors embedded in the context of daily life with the aim of identifying human behaviors, thoughts, feelings, and traits, with research showing particular promise for smartphone-based detection of depression, anxiety, and other mental health conditions.

CrossCheck [8] pioneered passive sensing for schizophrenia patients, using smartphone sensors to predict symptom trajectories without requiring active user input. The StudentLife project demonstrated similar capabilities for monitoring college student mental health through smartphone usage patterns. However, these sensing-focused approaches primarily address detection and monitoring rather than sustainable user engagement with interventions.

Just-in-Time Adaptive Interventions

The emergence of JITAI represents a significant advancement in mobile health interventions. JITAI employ adaptation to operationalize the provision of just-in-time support, providing the right type or amount of support at the right time, making them particularly relevant for mental health contexts where individual states can change rapidly and unpredictably.

Current JITAI research focuses heavily on physical activity and substance use, with limited application to broader mental health

contexts. The complexity of mental health symptoms and the vulnerability of clinical populations create unique challenges for JITAI implementation that require careful consideration of gamification design principles.

Implementation Gap in Digital Therapeutics

A comprehensive review by Sangiorgio [7] found that out of 2,300 reports on gamified health interventions, only 33 included meaningful measurements, with fewer than 10 showing statistically significant results. This measurement gap reflects deeper implementation challenges in clinical contexts, where regulatory constraints prevent the iterative optimization approaches that make consumer apps successful.

Implementation Context & Methods

Clinical Implementation Settings

Our implementation experience spans multiple clinical contexts, focusing on smartphone-first approaches that leverage ubiquitous computing principles without requiring additional hardware. At Sidekick Health, implementation targeted users with chronic illnesses across Scandinavia, Western Europe, and US cities, including a high-risk pregnancy app addressing postpartum mental health concerns for mothers on Medicaid. The platform integrated comprehensive health data through smartphone APIs, accessing step tracking, sleep patterns, and other personal informatics collected through Apple Health and Google Fit.

Work with Click Therapeutics focused specifically on cognitive remediation games for schizophrenia populations as part of ongoing FDA studies. These interventions required sophisticated adaptive difficulty scaling to accommodate extreme variability in cognitive capabilities within clinical populations. Additional consulting projects addressed addiction recovery apps, incorporating ethical safeguards to prevent exploitation of vulnerable users through manipulative gamification techniques.

Ubiquitous Computing Implementation Approach

Rather than requiring additional sensing hardware, our approach leveraged the ubiquitous smartphone platform that clinical populations already possess. This focus on accessible ubiquitous computing ensured scalability and reduced barriers to adoption, particularly critical for users in poverty who often lack predictable hardware access.

Technical implementation included dynamic difficulty scaling systems that tracked user performance metrics (error rates, response times, completion patterns) to provide personalized challenge levels. The adaptive step counter system demonstrated the practical application of JITAI principles, using smartphone-collected activity data to adjust daily goals based on individual capability and consistency patterns, resulting in a 20% improvement in user step achievement when properly calibrated.

Experiments with smartphone sensor integration included attempts to use microphone-based breathing detection for guided meditation exercises. However, technical limitations in detecting quiet breathing patterns led us to pivot toward more guided, less sensor-dependent approaches that maintained therapeutic value while ensuring consistent functionality across diverse hardware configurations.

Research Methodology

User research methods included surveys, focus groups, depth interviews, and UX research workshops, adapted for vulnerable populations through careful attention to informed consent and ethical considerations. Participant recruitment utilized specialized medical recruiting firms and university partnerships with established clinical populations, ensuring appropriate representation of target user groups.

Sample sizes followed industry best practices for UX research (qualitative interviews $n=10$ per project, quantitative surveys $n=70-100$) with broader user research across implementations reaching $n=300$ total participants. When working with digital therapeutics companies, research followed FDA standards and IRB protocols given clinical validation requirements. Churn analysis provided additional behavioral insights, with significant user dropoff serving as evidence of engagement challenges specific to clinical populations.

Data collection incorporated both traditional engagement metrics (retention at 1, 7, and 30 days, feature usage patterns, gamification point accumulation) and therapeutic-focused measures aligned with clinical outcome requirements. Analysis methods combined qualitative coding of user feedback with quantitative assessment of behavioral change indicators.

Key Findings & Adaptations

Social Mechanisms Cause Anxiety in Clinical Populations

User research across chronic illness populations consistently revealed that social mechanisms ranked dead last among motivational features, contradicting fundamental assumptions about social gamification effectiveness. A survey of approximately 100 users with chronic illness found social notifications (“you have a message waiting,” “your team needs help,” “share your success”) were consistently rated as least desirable features.

This finding aligns with broader research on personal informatics in mental health contexts, where health management, even when self-driven, is inherently social and depends on a person’s direct relationships and broader sociocultural contexts. However, clinical populations expressed strong preferences for consumption-only social features (reading relatable stories, gathering information) while actively avoiding exposure of their own health journeys to others.

Recent mothers recruited via Instagram for our postpartum mental health intervention showed similar patterns, expressing interest in accessing community knowledge while maintaining privacy around their personal health struggles. This pattern held across schizophrenia, addiction, and other clinical populations, requiring fundamental reconsideration of social gamification in clinical contexts.

Achievement Systems Require Population-Specific Calibration

Clinical populations respond differently to achievement mechanics than general users, necessitating careful adaptation of traditional gamification approaches. Achievement design principles emphasize that users must genuinely want rewards, feel ownership over them, and invest meaningful effort to earn them [8]. However, when people suffer from mental health conditions, they often lack the energy to pursue traditional achievement systems or may be triggered by competitive elements.

Research with schizophrenia patients revealed extreme variability in capabilities, requiring highly adaptive systems that could match individual skill levels while maintaining engagement. Our cognitive remediation games implemented threshold-based difficulty scaling: users performing at threshold maintained current difficulty, those below threshold moved to easier levels, and those above advanced upward. This approach required building extensive content libraries at each difficulty level to accommodate unpredictable user progression patterns.

Poorly designed achievement systems consistently felt patronizing when users were rewarded for actions they would take anyway, or demotivating when users felt challenged to do something their condition prevented. The key insight was matching challenge level to individual capacity while avoiding comparisons that highlighted limitations rather than progress.

Transparency Becomes Critical for Vulnerable Users

Vulnerable users require significantly clearer understanding of how gamification systems work compared to general populations. Clinical populations, already dealing with reduced agency due to their conditions, particularly value systems that clearly explain outcomes and provide predictable paths to desired results.

This finding connects to broader research on personal informatics systems, where transparency enables user control and agency. Opaque reward systems made users feel like objects of manipulation rather than active participants in their own care. Effective clinical gamification required explicit explanation of reward mechanisms, clear paths to achievement, and user control over feature engagement levels.

Smartphone-First Adaptive Systems

Our experience implementing adaptive interventions demonstrated the practical application of JITAI principles through

smartphone-based systems. The adaptive step counter exemplified this approach: initial simple goal escalation (increasing targets when met) created user frustration when exceptional activity days (vacation walking, hiking) dramatically increased subsequent goals without reverting.

By implementing more sophisticated adaptation algorithms—reverting goals downward when appropriate and only increasing for consistent progress patterns—we achieved a 20% improvement in user step achievement. While not a direct mental health intervention, physical activity such as daily steps interacts holistically with other mental health treatments. This demonstrates how ubiquitous computing principles can be applied through existing smartphone infrastructure to create effective adaptive interventions for clinical populations.

Integration with platform-level health data (Apple Health, Google Fit) provided comprehensive behavioral context without requiring additional user input, supporting JITAI decision-making through passive data collection. This approach addressed the accessibility constraints faced by clinical populations while maintaining the sensing capabilities necessary for adaptive intervention delivery.

Population-Specific Variability Challenges

Implementation with diverse clinical populations revealed significant heterogeneity that challenges assumptions about designing for clinical users as homogeneous groups. Schizophrenia patients showed extreme variability in cognitive capabilities: highly functioning users solved cognitive remediation puzzles quickly and needed significant challenge to stay engaged, while low-functioning users struggled with basic puzzles and became frustrated when difficulty scaled too rapidly.

This variability necessitated the development of sophisticated adaptive systems capable of real-time performance assessment and adjustment. Our threshold-based approach provided a practical framework for managing this complexity while maintaining therapeutic value across the full spectrum of user capabilities.

Framework Adaptations for Clinical Context

Rather than modifying established frameworks directly, we developed a reverse application approach that begins with deep understanding of specific user populations before selecting appropriate gamification mechanics. Through interviews and research with clinical populations, we identified which core drives motivated specific user groups, then selected mechanics that aligned with those drives while avoiding potentially harmful elements.

This approach proved particularly valuable for addiction populations, where traditional engagement mechanics from free-to-play mobile games could potentially exploit vulnerable users. By implementing usage limits and avoiding variable reward schedules that mirror addictive patterns, we maintained therapeutic benefit while ensuring ethical implementation.

The adaptation process emphasized matching gamification complexity to user capabilities and desires. Some users wanted simple acknowledgment and progress tracking, while others craved depth and optimization opportunities. Forcing inappropriate complexity levels created resistance rather than engagement, highlighting the need for user-configurable gamification intensity.

Recommendations & Future Work

Researchers developing gamified mental health interventions using ubiquitous computing technologies should follow a three-stage process: Research (understand the clinical context and work with subject matter experts), Engagement (conduct direct user research to understand genuine motivations within specific populations), and Design (adapt proven frameworks while testing extensively with target populations).

Key warning signs include users reporting anxiety from social features, achievement systems that feel patronizing or impossible, and opaque reward structures that reduce user agency. Effective collaboration with clinicians should focus on understanding clinical needs while educating healthcare providers about gamification's potential benefits and implementation risks.

Measurement should prioritize therapeutic outcomes alongside engagement metrics, with particular attention to user sense of control and system transparency. The smartphone-first approach to ubiquitous mental health support offers a scalable pathway for reaching clinical populations without the hardware barriers that limit accessibility of more complex sensing systems.

Future research needs include long-term habit formation studies, mechanism-specific research for particular clinical populations, and careful investigation of how “addictive” game mechanics interact with vulnerable users. The field faces significant implementation barriers, particularly in digital therapeutics where regulatory constraints prevent iterative improvement common in consumer app development.

Conclusion

The convergence of ubiquitous computing sensing capabilities with gamification expertise offers significant potential for addressing the engagement gap in digital mental health interventions. However, realizing this potential requires careful adaptation of established frameworks rather than direct application of consumer app strategies.

As sensing technologies become increasingly capable of detecting when mental health intervention is needed, gamification expertise becomes essential for bridging the gap between detection and sustained user engagement. Our smartphone-first approach demonstrates how ubiquitous computing principles can be applied through accessible technology platforms to create effective, scalable interventions for clinical populations.

The path forward demands respect for user agency, deep understanding of clinical contexts, and careful adaptation of

proven frameworks informed by rigorous user research with target populations. When implemented thoughtfully, gamification transforms digital mental health tools from compliance burdens into engaging partners in recovery and wellness, making the promise of ubiquitous mental health support a practical reality.

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